TECHFEST NER CHALLENGE | PAKISTANI CONTEXT

Named Entity Recognition for Pakistani Text

Fine-tuning XLM-RoBERTa on 20 entity types with self-collected, self-annotated data

Final project report

NEDUET | Dory's Data Dive | Round 2

Submitted: 21 October 2026

Team: _____ (fill in names)

Final model: xlm-roberta-base fine-tuned for token classification (41 BIO tags)

Contents

1. Executive summary 3

2. Problem statement and objectives 4

3. Approach overview 5

4. Model choice 5

5. Data collection and legal policy 6

6. Synthetic data 7

7. Annotation process 8

8. Dataset construction and leakage prevention 10

9. Model training 11

10. Evaluation protocol 11

11. Results 12

12. Urdu extension (zero-shot) 16

13. Challenges and how we solved them 16

14. Limitations 17

15. Reproducibility and repository 17

16. Conclusion and next steps 18

Appendix A. Licences and credits 19

Appendix B. Definitions 19

1. Executive summary

We built a named entity recognition (NER) system for Pakistani text that recognises the 20 entity types fixed by the TechFest challenge. We fine-tuned **XLM-RoBERTa-base** on a training set we collected and labelled ourselves from public sources (Wikipedia, Wikinews and openly available Kaggle datasets of Pakistani news and Supreme Court judgments), extended with synthetic sentences for the two types that must not use real personal data (phone numbers and CNICs).

0.814	0.816	0.812	20 / 20
micro F1, pooled human-checked test (764 entities)	F1 on random test (200 sentences)	F1 on weak-label test (100 sentences)	entity types handled

What we did. We (1) wrote annotation guidelines for all 20 types, (2) gathered about 15,400 clean sentences, (3) produced first-draft labels with regular expressions, gazetteers and a pretrained spaCy model, (4) corrected 500 sentences by hand to obtain gold test data, (5) learned the drafts' systematic errors and fixed them with rules, (6) built the training set with strict, automatically checked leakage controls, (7) fine-tuned the model, and (8) analysed errors and ran two improvement rounds.

Result. The final model scores **precision 0.806, recall 0.822, F1 0.814** (exact entity match) on 764 human-verified entities, and 14 of the 20 entity types reach F1 of at least 0.80. Phone numbers and CNICs, tested only on a held-out synthetic set, reach F1 0.990 and 0.918. The weakest types are ORGANIZATION (F1 0.392), LAW_ACT (0.452) and LANDMARK_FACILITY (0.564).

> **Honest finding**
> Two improvement rounds (more synthetic data and case-number rules; then masking of unlabelled names) changed the *kind* of error but did not raise overall F1 beyond run-to-run noise (all three runs score 0.810 to 0.814). We therefore selected the final model with a rule fixed in advance, not by the best single number. We report this openly because the limit is the amount of human-verified labels, not the model.

Also tested. The model was trained on English only, yet on a small hand-made Urdu sample it found 21 of 27 expected entities with the right type (zero-shot, tiny sample, see section 12).

2. Problem statement and objectives

The TechFest NER Challenge asks each team to choose a pretrained model, collect and annotate its own training data from public Pakistani sources, fine-tune the model on a fixed set of 20 entity types and report precision, recall and F1 (overall and per entity) on a benchmark that does not overlap with the training data. On 22 October all submissions are scored on a hidden benchmark prepared by the organisers. Phone numbers and ID numbers must be synthetic, scraping must respect website terms, and private or personal data must be avoided.

Entity types

Entity type	Covers	Entity type	Covers
PERSON	people's names	**LAW_ACT**	acts, articles, sections
CITY	cities, towns, districts	**EVENT**	tournaments, elections, days
PROVINCE_REGION	provinces and named regions	**MEDIA_OUTLET**	newspapers, channels
COUNTRY	countries	**SPORTS_TEAM**	teams incl. national sides
LANDMARK_FACILITY	landmarks, buildings, rivers	**DATE**	dates, ranges, durations
GOVERNMENT_BODY	ministries, police, agencies	**TIME**	time of day
POLITICAL_PARTY	parties	**MONEY**	amounts with currency (PKR)
ORGANIZATION	companies, NGOs, others	**PERCENT**	percentages
EDUCATIONAL_INSTITUTION	universities, schools	**PHONE_NUMBER**	phone numbers (synthetic)
COURT_LEGAL_BODY	courts, tribunals	**ID_NUMBER**	CNIC-format numbers (synthetic)

Our objectives, in priority order: (1) a legal, ethical and well-documented data pipeline; (2) honest evaluation without data leakage; (3) strong overall F1 and balanced per-entity scores; (4) a reproducible, explainable pipeline we can present; (5) optional extension to Urdu.

How NER is framed

Every word receives one tag: `B-X` (first word of an entity of type X), `I-X` (a continuation) or `O` (not an entity). Twenty types give 20 x 2 + 1 = **41 tags**. NER is therefore a token-classification problem. Entities are flat (no nesting) and the longest sensible span wins. An entity is correct only if its start, end **and** type all match (exact match), which is the strictest common scoring rule.

3. Approach overview

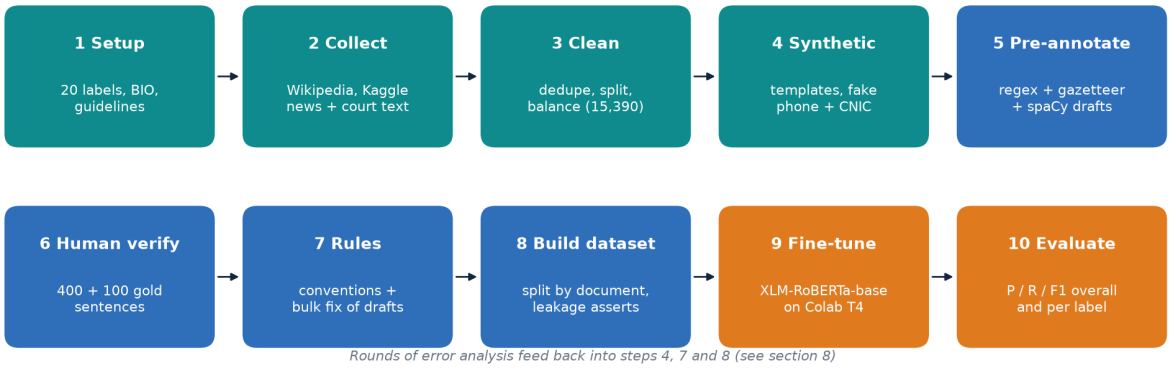


Figure 1. The ten-step pipeline. Colours: teal = data collection, blue = labelling and dataset building, orange = model and evaluation.

The central design idea is a **quality ladder** of labels. Gold labels (checked by a human) are few but trusted and used for testing. Silver labels (made by rules and a pretrained model) are plentiful but noisy and used for training, after the noisiest parts were fixed or masked. Synthetic labels are exact but artificial and cover what real text cannot (fake phone numbers and CNICs) or lacks (rare types). Everything downstream (splits, masking, evaluation) exists to keep this ladder honest.

4. Model choice

We chose **xlm-roberta-base** (270 million parameters, pretrained on text in 100 languages) with a linear classification layer for token tagging.

Candidate	Verdict	Why
XLM-RoBERTa-base	**Chosen**	Strong NER baseline; 250k sentencepiece vocabulary splits Pakistani and Urdu-origin names sensibly; multilingual so Urdu is possible without changing the model; trains in minutes on a free Colab T4 GPU.
BERT-base / RoBERTa-base	Rejected	English-only vocabularies handle unfamiliar names and Urdu-origin words worse.
mBERT	Rejected	Older multilingual model, generally weaker than XLM-R on NER.
DeBERTa-v3-base	Rejected	Slightly stronger on English, but slower, with tokenizer pitfalls on Colab; not worth the risk on a short deadline.
spaCy NER	Used only as a labeller	Fast, but lower accuracy, and it was our noisy draft labeller; copying it would add no skill.
LLM prompting	Not used as the system	The task requires a fine-tuned model.
XLM-RoBERTa-large	Rejected	Needs more memory and time; little gain with noisy labels.

5. Data collection and legal policy

Our policy ranks sources by legal risk: openly licensed text first, official public records second, commercial news kept small. We checked terms for each source, rate-limited requests, identified our scraper with a User-Agent that contains a contact address, and recorded every source in `DATA_SOURCES.md`. We use text only to train and evaluate the model and we do not redistribute the dataset.

Source	What we used	Terms / note
Wikipedia (official API)	209 articles on Pakistan	CC BY-SA 4.0; credited here; 1.5 s delay between requests
Wikinews (official API)	23 articles	CC BY 2.5
Kaggle: Pakistan news headlines	about 26.7k headlines, Pakistan-related favoured	public dataset
Kaggle: Dawn and Express Tribune articles	778 + 3,309 articles (2013 and 2018)	public dataset; short snippets only in this report
Kaggle: Supreme Court of Pakistan judgments	2,809 judgments (public records)	public court records; headers and citation lists removed
Kaggle: Pakistan Penal Code, census table	gazetteers (law sections, provinces, districts)	public dataset

Removed after inspection. We dropped Imran Khan tweets (personal social media and garbled Urdu encoding), a tweet set on India-Pakistan cricket (informal, not Pakistan-specific, four duplicate formats) and an Imran Khan Reddit set (informal forum posts). No private or personal data is used. Phone numbers and CNICs are generated randomly, never taken from a person.

Cleaning and balancing

Raw text became **15,390 clean sentences** (`data/raw/sentences.jsonl`). The cleaning steps and the reason for each:

- **Tolerant decoding:** files mix UTF-8 with stray cp1252 bytes; a fallback decoder avoids crashes and garbled characters.
- **Normalisation:** fancy quotes, dashes and non-breaking spaces become plain characters, so the tokenizer sees consistent input.
- **Junk removal:** photo captions, "Read more" lines, URLs, ALL-CAPS headers, very short or very long lines and OCR-broken sentences are not natural language and would teach noise.
- **Abbreviation-safe sentence splitting** (Mr., Rs., M. Khan) so entities are not cut in half.
- **Court text:** header blocks (judge lists, case numbers) and citation lists were cut.
- **De-duplication:** exact and near-duplicate removal (same first 12 words).
- **Entity-light sentences** (no capitalised word or digit) were mostly dropped (25% down to 11%); a small share remains as negative examples.
- **Balancing:** caps per source and per document so 65 MB of legal text does not dominate. Result: 18% open-licence text, 24% public records, 58% news.
- **Document id:** every sentence keeps its source document (`group`), which makes leakage-safe splitting possible later.

6. Synthetic data

Phone numbers and CNICs must be synthetic by the rules, and some entity types are rare in real text. We generated **5,390 sentences** (6,260 in later runs) from about 150 sentence templates with slots such as `{PERSON}`, `{CNIC}` and `{MONEY}`. The static text and each slot are tokenised separately and entity tokens receive B-/I- tags, so the labels are exact with no manual work.

- **CNIC:** `#####-#######-#` (first digit 1-7) plus no-dash and spaced variants. **Phone:** mobile `0300-1234567`, `+92 300 1234567`, `+923001234567`, `0092-...` and landlines. All digits are random.
- **MONEY, PERCENT, DATE, TIME:** many formats (Rs, Rs., PKR, $, lakh, crore, million, Tribune-style Rs5.63 million; 15%, 15 percent, 15 per cent).
- **Rare types** (SPORTS_TEAM, MEDIA_OUTLET, EVENT, LANDMARK_FACILITY, LAW_ACT, COURT_LEGAL_BODY, EDUCATIONAL_INSTITUTION) get 260-320 sentences each.
- **Names** are random first-name and surname combinations, not real individuals.
- **Safeguards:** templates include unlabelled look-alike numbers (FIR numbers, flight numbers, page counts) so the model does not learn that every number is an entity. The tokenizer is shared with real data, so numbers such as 35202-1234567-1 and 1,200 stay single tokens.

Risk we accept: the model may learn template wording. We keep synthetic data a minority (5,390 against 15,390 real sentences) and test the phone and CNIC patterns on a separate held-out synthetic set with new templates, unseen formats and decoys (section 11.5).

7. Annotation process

Hand-labelling 15,000 sentences in three days is impossible, so we used **pre-annotation**: automatic first drafts that humans correct. Correcting is far faster than labelling from scratch.

7.1 Automatic drafts (silver labels)

The draft combines three layers, and every span remembers which layer proposed it: (1) **regular expressions** for phone, CNIC, money, percent, time, date and law references; (2) **gazetteers** (curated Pakistani lists of provinces, divisions, districts, courts, parties, abbreviations) matched longest-first; (3) **spaCy's pretrained NER** for persons and generic organisations, re-classified by keyword rules (University becomes EDUCATIONAL_INSTITUTION, Court becomes COURT_LEGAL_BODY, Ministry and Police become GOVERNMENT_BODY). Overlaps are resolved by strong regex first, then longest span, then layer priority. Three review rounds fixed systematic mistakes we saw, for example a newspaper tagged as a person, a mosque as a person, party-name halves as cities, bare "Court" tagged 385 times, and countries as cities. Each fix was checked with unit cases.

7.2 Human verification and gold test data

We hand-checked **400 sentences in Label Studio**: 200 **random test** sentences (never trained on) and 200 **review** sentences (half rare-label coverage, half uncertain cases; used for training and for learning the drafts' mistakes). An AI assistant first corrected the drafts to speed up the work, and the human then reviewed every sentence. We disclose this: the labels are model-assisted and human-verified, and it is the human pass that makes the test set gold. We then verified another **100 sentences (extra_test)** chosen to be rich in weak types. Because extra_test is selected, not random, it is always reported separately.

How good were the automatic drafts?

Measured against the human labels, the automatic draft was strong for PERSON (about 0.88), POLITICAL_PARTY (about 0.9), MONEY and PERCENT, but weak for ORGANIZATION (precision about 0.3), GOVERNMENT_BODY (recall about 0.57) and LANDMARK, EVENT and SPORTS_TEAM (recall 0.3 to 0.5). The assistant-corrected version reached about 0.92 precision and 0.89 recall on the 100-sentence sample. This evidence drove every later decision.

7.3 Annotation conventions

Rules that were found while annotating were written into `GUIDELINES.md` and applied automatically, because the model learns whatever is consistent:

- Durations and periods ("two weeks", "23 months") are DATE (OntoNotes convention); TIME is time of day only. "Thursday morning" is two spans: DATE plus TIME.
- A year inside a name stays inside it ("Elections Act 2017" is one LAW_ACT); a year inside a case number or citation (C.A.538/2022, 2010 SCMR 1962) is never a DATE.
- Leading "The" is not part of a name. Courtesy titles (Mr, Dr, Justice, Prime Minister) are not part of PERSON.
- Capitalised bare court names ("High Court", "Supreme Court") are COURT_LEGAL_BODY because the hidden benchmark is likely court-record text; generic "the court" is not labelled.
- Kashmir, Thar and similar regions are PROVINCE_REGION; districts and towns are CITY; rivers, stadiums, police stations, bazaars and shrines are LANDMARK_FACILITY.
- National teams in a match context (Pakistan, England) are SPORTS_TEAM; otherwise COUNTRY. Abbreviations take the type of what they stand for (WAPDA is GOVERNMENT_BODY).
- FIR, flight and case numbers and page counts are not ID_NUMBER or PHONE_NUMBER.

7.4 Bulk fix of the silver labels

A human fixes one sentence; a *pattern* of errors can be fixed everywhere. We measured draft precision for every (label, layer) pair on the review split only, then applied rules to all 14,990 non-human sentences: added missing gazetteer names, dropped junk (IGP, CEC, "Local Commission"), kept TIME only if it has a digit or time-of-day word, split "Wednesday night", folded years into law and event names, and converted national teams to SPORTS_TEAM when strong sport cues are in the sentence (a first, broader cue list was caught by spot-checking). Sentences with a remaining unreliable spaCy span (ORGANIZATION precision 0.32, GOVERNMENT_BODY 0.43, TIME 0.20) are flagged *untrusted* (11,295 trusted, 3,695 untrusted). The flag separates quality well: exact-match agreement with the human is 0.68 to 0.72 for trusted and 0.04 to 0.32 for untrusted sentences. The test set was never used to tune any rule.

7.5 Weak-class patch

Label counts showed LANDMARK_FACILITY, ORGANIZATION, EDUCATIONAL_INSTITUTION, EVENT, TIME and SPORTS_TEAM were thin, because the trust rule drops sentences with unreliable spans, which are mostly these labels. We selected 300 extra sentences (one per document, round-robin over the weak labels): 100 became the human-verified **extra_test** and 200 became **extra_train** (labelled by the assistant, *not* human-verified, and disclosed as such). extra_test documents are disjoint from all other documents, asserted in code.

8. Dataset construction and leakage prevention

Data leakage means the model sees in training something that is in the test. Then the score measures memory, not skill. Sentences from the same document share names and wording, so splitting by sentence would leak. We split by **document** and verify every rule with assertions that stop the build if anything leaks.

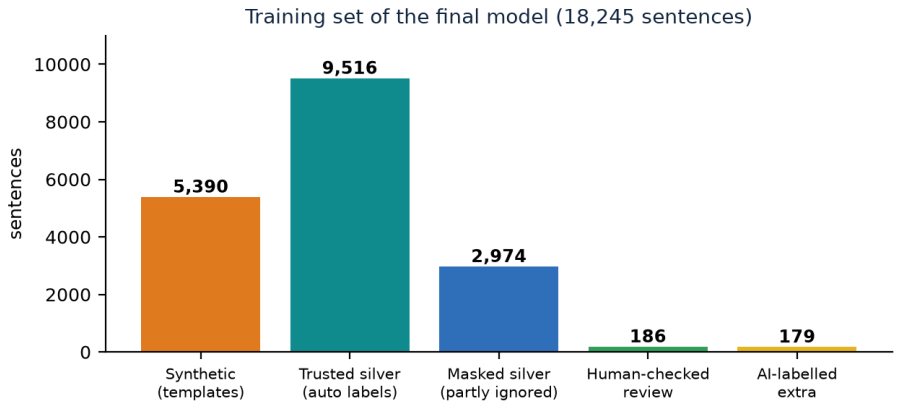


Figure 2. Composition of the final model's training set.

Control	Result
Whole test documents removed from training	965 silver sibling sentences dropped (196 + 100 test documents)
Silver copies of human or assistant sentences	300 dropped; the human version is used
Near-duplicate check against both test sets (first 12 words, shared 8-word run)	64 training sentences dropped
Validation set	8% of document groups (seed 42), separate from test, used only to choose the best epoch
Synthetic data	training only
Hard assertions in the build script	disjoint document groups, no test id or text in train, tags valid, lengths match: all passed

Masking instead of dropping. Silver spans from unreliable layers (spaCy ORGANIZATION, GOVERNMENT_BODY, EDUCATIONAL_INSTITUTION, EVENT, LANDMARK and MONEY) keep their words but get label `-100`, which the loss function ignores. Dropping the sentence would lose good labels; keeping the label would teach false facts; leaving the words as "O" would teach the model that a real entity is not an entity. Masking removes the noisy signal and keeps the rest.

Split	Sentences	Origin
Train (final model)	18,245	synthetic 5,390; trusted silver 9,516; masked silver 2,974; human review 186; assistant extra 179
Validation	1,206	document-level split of the same pool
Test (gold)	200	random, human-verified, 426 entities, 196 documents
Extra test (gold)	100	weak-label enriched, human-verified, 338 entities
Pattern test (synthetic)	188	new templates, unseen formats, decoys: 102 phones, 84 CNICs

9. Model training

How fine-tuning works. The model reads batches of 16 sentences. Each word is cut into subword pieces; a new linear layer turns the first piece of every word into a score for each of the 41 tags. Cross-entropy loss measures the gap to the target tags (ignoring `-100` positions: special tokens, later pieces of a word and masked spans), back-propagation computes how each weight should change and the AdamW optimiser applies a small update: one step. One pass over the data is one epoch. After every epoch the model is scored on validation data (not trained on) and saved; the epoch with the best validation F1 is kept. Only then are the test sets scored, once.

Setting	Value	Why
Model	xlm-roberta-base + linear tag layer	section 4
Epochs (final model)	4	standard for NER fine-tuning; best epoch kept on validation F1
Learning rate	3e-5, 10% warm-up then linear decay	protects pretrained knowledge at the start; settles at the end
Batch size	16	fits a 16 GB T4
Weight decay / precision	0.01 / fp16	mild regularisation / faster, less memory
Max length	192 word pieces	one training sentence was truncated
Oversampling	human review x3, assistant extras x2	the best data is only about 2% of the total
Labelling rule	first sub-token carries the label	other pieces and masked spans = -100
Seed	42	reproducible

Training ran on Google Colab (NVIDIA Tesla T4, 16 GB) and took a few minutes per run. Libraries: transformers 5.20.2, accelerate 1.15.0, torch 2.11.0+cu130, numpy 2.1.3. We implemented the entity-level metric ourselves after the seqeval package failed to build on Colab, and we avoided the Hugging Face `datasets` package because our project folder of that name would shadow it.

10. Evaluation protocol

- **Metric:** entity-level exact match. A predicted entity is correct only if start, end and type all match. Precision = correct / predicted; recall = correct / true; F1 is their harmonic mean. Micro scores pool all entities; macro scores average the labels.
- **Sets:** random test (200 sentences, representative), extra_test (100 sentences, weak-label enriched), pattern_test (synthetic, phone and CNIC). The two human sets are also **pooled (764 entities)** because many labels have only 2 to 15 examples per set, which makes single-set scores noisy.
- **No tuning on test:** validation data selects the epoch; test sets were scored only after training.
- **Run-to-run noise:** with sets this small, differences below about 0.02 F1 are noise, so we do not claim one run is better unless it clears that margin.

11. Results

11.1 Final model: overall scores

Evaluation set	Sentences	Entities	Precision	Recall	F1
Random test (gold)	200	426	0.805	0.826	**0.816**
Extra test (gold, weak-label enriched)	100	338	0.807	0.817	**0.812**
Pooled gold (both sets)	300	764	**0.806**	**0.822**	**0.814**
Pattern test (synthetic, PHONE + ID only)	188	186	see 11.5	see 11.5	see 11.5

The two human sets give almost the same F1 (0.816 and 0.812) although extra_test is deliberately harder. A model that had only memorised easy patterns would drop on the harder set, so this consistency is evidence of real generalisation.

11.2 Per entity type

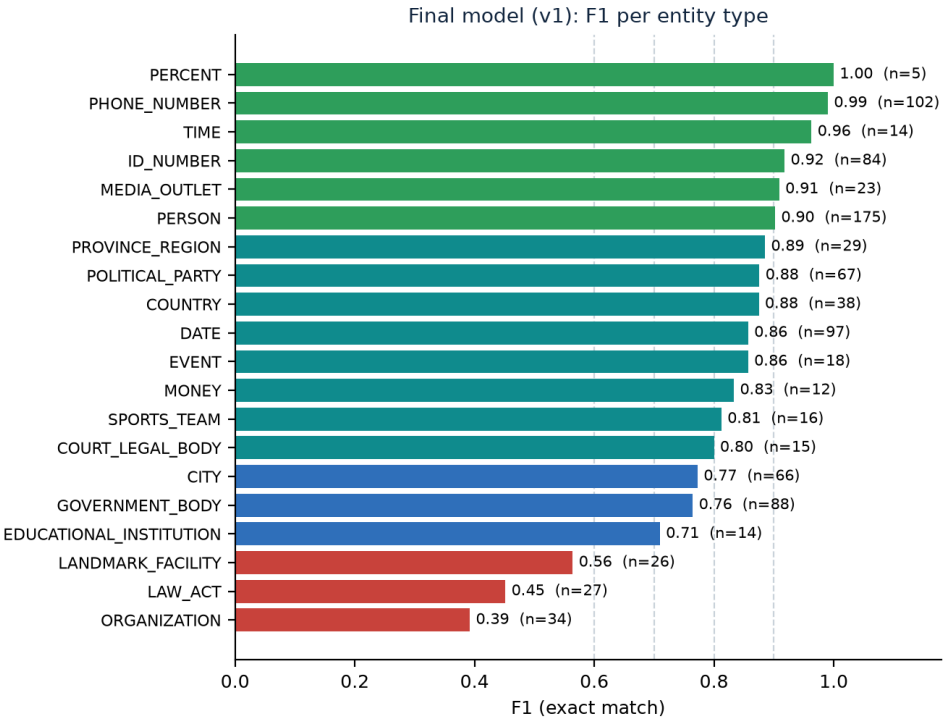


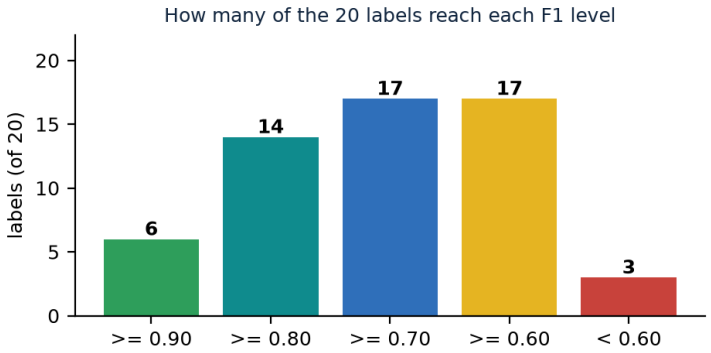
Figure 3. Pooled F1 per entity type for the final model (n = number of gold entities; PHONE_NUMBER and ID_NUMBER from the synthetic pattern test).

Entity type	Test P / R / F1	Extra-test P / R / F1	Pooled F1	n
PERSON	0.89 / 0.94 / 0.92 (122)	0.80 / 0.96 / 0.87 (53)	0.902	175
CITY	0.78 / 0.74 / 0.76 (34)	0.76 / 0.81 / 0.79 (32)	0.773	66
PROVINCE_REGION	0.77 / 1.00 / 0.87 (17)	1.00 / 0.83 / 0.91 (12)	0.885	29
COUNTRY	0.81 / 0.93 / 0.86 (27)	0.91 / 0.91 / 0.91 (11)	0.875	38
LANDMARK_FACILITY	0.67 / 0.25 / 0.36 (8)	0.90 / 0.50 / 0.64 (18)	0.564	26
GOVERNMENT_BODY	0.75 / 0.68 / 0.71 (53)	0.85 / 0.83 / 0.84 (35)	0.765	88
POLITICAL_PARTY	0.85 / 0.92 / 0.88 (36)	0.87 / 0.87 / 0.87 (31)	0.876	67
ORGANIZATION	0.64 / 0.35 / 0.45 (20)	0.50 / 0.21 / 0.30 (14)	0.392	34
EDUCATIONAL_INSTITUTION	1.00 / 1.00 / 1.00 (2)	0.60 / 0.75 / 0.67 (12)	0.710	14

Entity type	Test P / R / F1	Extra-test P / R / F1	Pooled F1	n
COURT_LEGAL_BODY	0.82 / 0.90 / 0.86 (10)	0.75 / 0.60 / 0.67 (5)	0.800	15
LAW_ACT	0.41 / 0.58 / 0.48 (12)	0.39 / 0.47 / 0.42 (15)	0.452	27
EVENT	1.00 / 1.00 / 1.00 (3)	0.86 / 0.80 / 0.83 (15)	0.857	18
MEDIA_OUTLET	1.00 / 0.83 / 0.91 (6)	0.94 / 0.88 / 0.91 (17)	0.909	23
SPORTS_TEAM	0.33 / 0.33 / 0.33 (3)	0.92 / 0.92 / 0.92 (13)	0.812	16
DATE	0.76 / 0.88 / 0.81 (57)	0.85 / 1.00 / 0.92 (40)	0.857	97
TIME	1.00 / 1.00 / 1.00 (2)	1.00 / 0.92 / 0.96 (12)	0.963	14
MONEY	1.00 / 0.89 / 0.94 (9)	0.50 / 0.67 / 0.57 (3)	0.833	12
PERCENT	1.00 / 1.00 / 1.00 (5)	n/a	1.000	5
PHONE_NUMBER	synthetic only	0.98 / 1.00 / 0.99 (102)	0.990	102
ID_NUMBER	synthetic only	0.85 / 1.00 / 0.92 (84)	0.918	84

The phone and ID rows come from the synthetic pattern test, so they are optimistic; no real phone or CNIC text exists by design.

11.3 How many types reach each score level



Of the 20 types, 6 reach F1 >= 0.90, 14 reach >= 0.80, 17 reach >= 0.70, and 3 stay below 0.60 (ORGANIZATION, LAW_ACT, LANDMARK_FACILITY). Supports differ widely (PERSON n=175, PERCENT n=5), so a label near a band edge can change band from one or two entities; the bands are descriptive, not a ranking.

11.4 What the errors look like

Every wrong word of the first model was classed as **missed** (gold entity, model said O), **spurious** (gold O, model said entity), **confusion** (wrong type) or **boundary** (right type, wrong edges). Each kind has a different cure: missed means too few examples, spurious means noisy training labels, confusion means unclear type borders, boundary means inconsistent span rules.

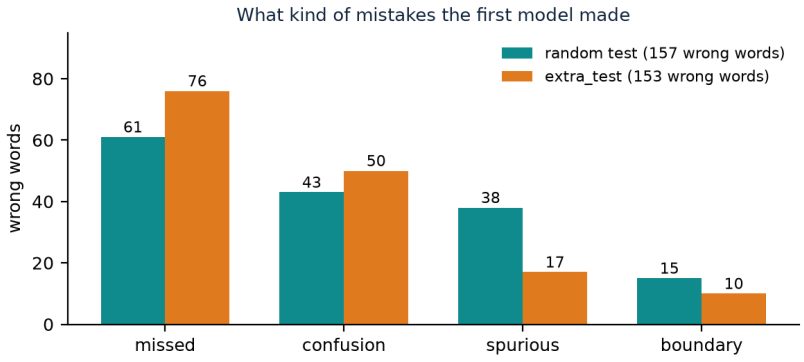


Figure 4. Error kinds of the first trained model (v1, before any improvement round).

- **ORGANIZATION vs GOVERNMENT_BODY vs POLITICAL_PARTY** is the top confusion in every run (OGRA, CCP, PSB, SCBA were typical). Few clean examples exist, and the spaCy ORGANIZATION layer was masked.
- **LAW_ACT:** long names with brackets, years and "and" were split. Training laws were mostly short.
- **LANDMARK_FACILITY:** rivers, dams, ports, shrines, temples and police stations were missed or tagged CITY or PERSON.
- **Spurious DATE:** 18 of the 38 spurious words on the random test; many were years inside case numbers or citations.
- **COUNTRY vs SPORTS_TEAM:** "Pakistan" in a tour or match context is context-dependent.
- A few gold labels look questionable (for example one Kashmir tagged CITY). We never edited test labels.

11.5 Phone numbers and CNICs

No human test set contains a phone or CNIC (we may not use real personal data), so the first reports listed only 18 labels. We added a held-out **synthetic pattern test** of 188 sentences: new templates, about 30% unseen formats (such as `+92 (300) 1234567`, `0300.1234567`, `35202/1234567/1`) and look-alike decoys (IBANs, number plates, tracking ids, case references, 6-digit codes). Results for the final model:

Entity	Support	Precision	Recall	F1
PHONE_NUMBER	102	0.981	1.000	**0.990**
ID_NUMBER	84	0.848	1.000	**0.918**

The model finds every phone and CNIC, including unseen formats; the remaining weakness is false alarms on look-alikes (tracking ids and plates tagged as ID_NUMBER). This shows the model learned the patterns, but it is **not** evidence about real documents. An early version of this test contained date, time, percent and money decoys that the model (correctly) tagged; we found this flaw, removed those decoys and re-scored, and quote only the clean numbers.

11.6 Improvement rounds and model selection

Round 1 (v2): new rule that years in case numbers and citations are never DATE, 870 more synthetic sentences (long laws, landmarks, bodies, case-number noise) and 6 epochs. **Round 2 (v3):** in silver sentences, capitalised name-like tokens that the draft left as O (possible missed organisations and landmarks) were masked, so the model is not taught that they are not entities. On review data this rule covered 46% of tokens the draft had missed while masking only 0.7% of truly-O tokens. **Honesty rule:** because error *categories* were found on the test sets, those sets are no longer fully unseen; we added no test entity string to any data, validated rules on the review split and never edited test labels.

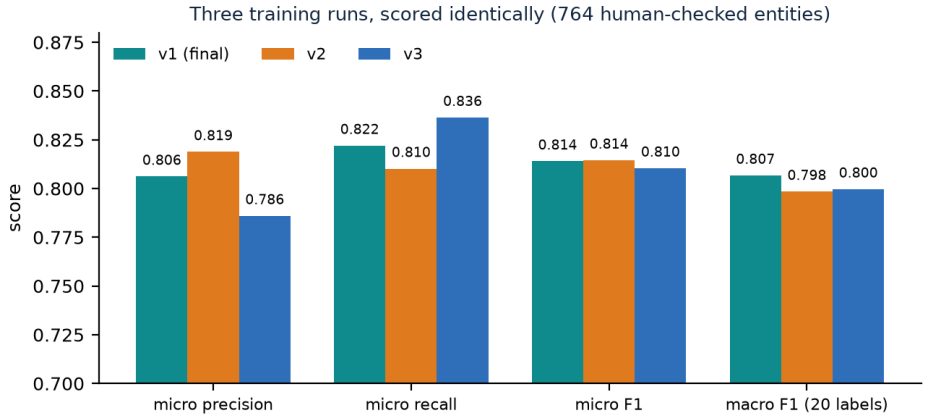


Figure 5. All three runs re-scored with the same code and data (pooled human gold; macro F1 over 20 labels).

Run	Micro P	Micro R	Micro F1	Macro F1 (20)	|P - R|
v1 (final): 4 epochs, first data	0.806	0.822	0.814	0.807	0.016
v2: + synthetic, case-number rule, 6 epochs	0.819	0.810	0.814	0.798	0.009
v3: + masking of unlabelled names	0.786	0.836	0.810	0.800	0.050

The runs differ by less than 0.005 in micro F1, far below the noise level. The rounds changed the error profile (v2 gained precision; v3 gained recall, with ORGANIZATION 0.45 and LANDMARK_FACILITY 0.65 best, but more spurious and wrong-type words) without a measurable overall gain, and LAW_ACT stayed near 0.45 in every run.

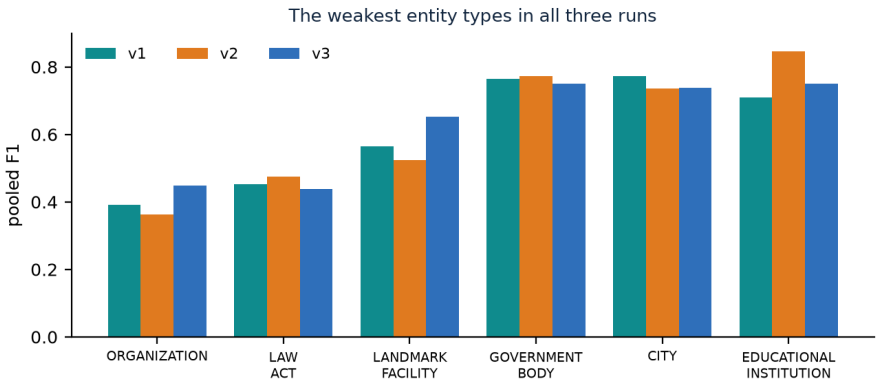


Figure 6. The weak types in all three runs.

Selection rule, fixed in advance
(1) pooled micro F1 over the two human test sets; (2) among tied runs, the best macro F1 over all 20 labels (18 human-tested plus PHONE_NUMBER and ID_NUMBER from the pattern test); (3) tie-break on the smallest precision-recall gap; (4) never choose on one set or one label. Step 1 is a tie (0.814 / 0.814 / 0.810); step 2 gives v1 0.807, v2 0.798, v3 0.800. **v1 is the final model**, which is also the simplest. We fixed this rule before re-scoring all three models with an evaluate-only mode (the v3 re-score reproduced the original numbers exactly).

A side-by-side run of all three models on one long paragraph containing all 20 types showed 50 of 54 predicted entities identical in every model. The differences were: v1 alone found "Dawn" and the full name "Anwaar-ul-Haq Kakar" (v2 cut it); v1 called "Karachi Press Club" a MEDIA_OUTLET (v2 and v3 correctly said LANDMARK_FACILITY); all three cut a long name ("Bilawal Bhutto Zardari") to two words.

12. Urdu extension (zero-shot)

The organisers noted that Urdu NER may earn extra points. Our model was trained on English only, but XLM-RoBERTa was pretrained on 100 languages including Urdu, so some Urdu entities may already be recognised. We ran the final model, unchanged, on **8 hand-written Urdu sentences** (27 expected entities covering 18 types, with fake phone and CNIC numbers) and compared the output with the entities we expected.

Result (exact match)	Value
Expected entities found with the right type	21 of 27
Precision / recall / F1	0.750 / 0.778 / 0.764
Types fully correct	PERSON 3/3, CITY 4/4, MEDIA_OUTLET 2/2, SPORTS_TEAM 2/2, PHONE_NUMBER, ID_NUMBER, MONEY, PERCENT, TIME, EVENT, GOVERNMENT_BODY, COURT_LEGAL_BODY, EDUCATIONAL_INSTITUTION, LANDMARK_FACILITY (1/1 each)
Not found or wrong	province names tagged CITY (2); DATE 0/2 because the Urdu postposition "ko" was included in the span; LAW_ACT cut to the year only; POLITICAL_PARTY split in two

Most misses are **boundary** errors caused by Urdu grammar (a postposition attached to a date), which targeted Urdu training data would fix. Our word tokenizer already splits Urdu correctly (the Urdu comma and full stop become separate tokens, numbers stay whole). **Caveat:** 27 entities written by us, not verified by a native annotator, no Urdu training: this is a sanity check, not a benchmark. We expect a small Urdu fine-tuning round (synthetic Urdu templates plus 60 to 100 hand-labelled Urdu sentences, mixed with the English data) to fix the boundary and province errors, but we did not run it, so we make no claim about its effect.

13. Challenges and how we solved them

Challenge	What happened	How we handled it
Few human labels	Only 500 sentences are human-verified out of 18,000+	Drafts + correction, bulk fix from review errors, masking, oversampling the best data
Noisy automatic labels	spaCy ORGANIZATION precision 0.32, TIME 0.20	Measured precision per layer, trusted flag, masked unreliable spans instead of dropping sentences
Data leakage	Same-document sentences share names and wording	Group-level splits, sibling removal, near-duplicate check, hard assertions
Legal and ethical sourcing	Competition forbids personal data and ToS violations	Source tiers, API access, rate limits, removed social-media sets, synthetic phone and CNIC
Ambiguous types	Court vs government body, team vs country, organisation vs government body	Written guidelines and automatic convention rules
Rare types	Random test has only 2-3 spans of EVENT, TIME, EDUCATIONAL_INSTITUTION	extra_test, pooled scoring, synthetic data
Encoding and OCR noise	Mixed UTF-8 and cp1252, broken court text	Tolerant decoder, OCR-sentence filters
Tooling failures	seqeval failed to build; transformers changed warm-up argument; folder name shadowed a library	Own metric; version-adaptive training arguments; plain torch dataset
Measuring phone and CNIC	No real data allowed, so reports showed 18 labels	Held-out synthetic pattern test with decoys and unseen formats; flawed first version found and fixed
Improvement rounds that did not help	Overall F1 stayed 0.81	Reported honestly; selected model by a pre-agreed rule

14. Limitations

- Most training labels are machine-made (silver). Human-verified data is about 500 sentences; the 179 extra sentences are assistant-labelled and not human-verified.
- Test sets are small. Per-label scores for rare types rest on 2 to 18 entities and are noisy; run-to-run noise is about 0.02 F1.
- Test sets are not fully unseen: error categories were found on them (no test string was added to data and no test label edited). The hidden benchmark is the clean measure.
- PHONE_NUMBER and ID_NUMBER are measured only on synthetic data, which resembles the training templates; real-document performance for these two types is unmeasured by design.
- Annotation conventions are judgement calls (for example durations as DATE, bare "High Court" as COURT_LEGAL_BODY) and the hidden benchmark may differ.
- Weak types: ORGANIZATION (0.39), LAW_ACT (0.45) and LANDMARK_FACILITY (0.56) are well below the rest.
- English only; Urdu is a zero-shot sanity check on 27 entities.

15. Reproducibility and repository

Everything runs from the project root. The pipeline order is: `scrape_wikipedia.py`, `build_corpus.py`, `generate_synthetic.py`, `preannotate.py`, Label Studio, `import_labelstudio.py`, `normalize_conventions.py`, `bulk_fix_silver.py`, `build_dataset.py`, then training in the Colab notebook. Prediction: `python scripts/predict.py --model outputs/xlmr_base/best --input file.txt --out predictions.jsonl` (entities with label, text and character offsets, plus tokens and BIO tags).

Path	Content
scripts/	all pipeline, training, evaluation, comparison and prediction code
data/dataset/	train, validation, test, extra_test, pattern_test (jsonl) and the leakage report
notebooks/	01 pre-annotation, 02 training, 03 evaluate and predict with all models, 04 Urdu check
trained_reports/	metrics, reports and error lists for every run
GUIDELINES.md, DATA_SOURCES.md	annotation rules and the source log
PROJECT_LOG.md, ERROR_ANALYSIS.md, README.md	decisions with numbers, error-analysis method, project overview

Seeds are fixed (42). Library versions are listed in section 9. Scores are produced by one code path (our own exact-match scorer) for all runs.

16. Conclusion and next steps

We delivered a complete, documented and leakage-controlled NER pipeline for Pakistani text. The final fine-tuned XLM-RoBERTa-base reaches F1 0.814 on 764 human-verified entities, with 14 of 20 types at F1 >= 0.80, near-perfect phone and CNIC pattern recognition on a held-out synthetic set and a promising zero-shot result on Urdu. Our most important finding is that label quality, not model size or training length, limits the score: three differently trained models tie within noise.

Next steps with more time: (1) verify the 200 assistant-labelled sentences and add human labels for ORGANIZATION, GOVERNMENT_BODY, LAW_ACT and LANDMARK_FACILITY; (2) add a gazetteer for education institutions; (3) run a small Urdu fine-tuning round with hand-labelled Urdu sentences; (4) average several seeds to shrink run-to-run noise; (5) adapt the output format once the organisers publish the benchmark format.

Appendix A. Licences and credits

- Wikipedia text: CC BY-SA 4.0 (credited). Wikinews: CC BY 2.5.
- Kaggle datasets (news headlines, Dawn and Express Tribune articles, Pakistan Penal Code, census table, Supreme Court judgments): public datasets used only to train and evaluate the model, not redistributed. Per-dataset licences were not individually verified.
- Model: xlm-roberta-base (Facebook AI), released under the MIT licence; accessed through Hugging Face transformers.
- Tools: Label Studio (annotation), spaCy (draft labels), PyTorch, Hugging Face transformers and accelerate, Google Colab (T4 GPU).
- An AI assistant helped draft first-pass corrections and write code; all test labels were verified by a human and this use is disclosed in section 7.

Appendix B. Definitions

Term	Meaning
Gold / silver / synthetic	human-verified labels / automatic labels with noise / template-generated labels
Precision, recall, F1	correct / predicted; correct / true; harmonic mean of the two
Exact match	an entity counts only if start, end and type all match
Micro / macro	pool all entities / average the per-label scores
BIO tags	B- first word, I- continuation, O outside
Masking (-100)	positions the loss function ignores
Data leakage	test information visible during training
Zero-shot	model applied to a language or task it was not trained on